Types of Operating System

Tasks

1. Create a table comparing at least three types of operating systems (such as Windows, Linux, and Android) based on their main features, target devices, and examples of apps you use on each.

Ans. :-

Operating System	Main Features	Target Devices	Examples of Apps
Windows	User-friendly interface, wide software support, gaming support, multitasking	Desktops, laptops, tablets	Microsoft Word, Chrome, VLC, Photoshop
Linux	Open-source, secure, customizable, stable, many distributions	Desktops, laptops, servers	Firefox, LibreOffice, VLC, Terminal
Android	Touch-friendly, app-based, customizable, supports mobile connectivity	Smartphones, tablets, smart TVs	WhatsApp, YouTube, Instagram, Google Maps
macOS	User-friendly, strong integration with Apple hardware, multimedia support	MacBook, iMac, Mac Studio	Safari, Final Cut Pro, Xcode, Pages
iOS	Secure, smooth interface, strong Apple ecosystem integration	iPhone	Safari, FaceTime, WhatsApp, Instagram

2. Identify the operating system running on your phone and your laptop/desktop. Write down two unique features of each OS that affect your daily use of popular apps like Instagram or Zomato.

Ans.:-

Device	Operating System	Two Unique Features	Effect on Daily Apps
Phone	**Android**	1. Multitasking & notifications 2. Google Play Store & app permissions	Instagram notifications can appear instantly, and apps like Zomato can use location permission for nearby restaurants and delivery tracking.
Laptop/Desktop	**Windows 11**	1. Multitasking with multiple windows 2. Wide software and browser support	You can use Instagram/Zomato through a web browser while working on other applications, such as Word or Chrome.

3. List three security risks that are specific to mobile operating systems like Android or iOS, and explain how these risks could impact apps you use (for example, WhatsApp or Paytm).

Ans.:-

Three Mobile Operating System Security Risks :-

Security Risk	Explanation	Impact on Apps
1. Malicious Apps	Apps downloaded from untrusted sources may contain malware or spyware.	A malicious app could steal WhatsApp messages, contacts, or sensitive information from Paytm.
2. Excessive App Permissions	An app may request unnecessary access to contacts, camera, microphone, location, or storage.	If permissions are misused, an app could access personal data or track your location.
3. Outdated OS / Security Updates	Older Android/iOS versions may contain security vulnerabilities that have already been fixed in newer updates.	Attackers could exploit vulnerabilities to access data or interfere with apps such as WhatsApp or Paytm.

4. Research and describe how a cloud-based operating system (such as Chrome OS or a virtual desktop environment) is different from a traditional desktop OS. Give one example of how this difference changes the way you access apps like YouTube or Flipkart.

Ans.:-

Cloud-Based OS vs Traditional Desktop OS :-

FEATURE	CLOUD-BASED OS	TRADITIONAL DESKTOP OS
MAIN IDEA	Relies heavily on cloud services and web applications.	Runs most applications and stores much of the data locally.
EXAMPLES	Chrome OS, virtual desktop environments	Windows, macOS, Linux
APPS	Many apps are accessed through a browser or cloud platform.	Applications can be installed directly on the computer.
STORAGE	Often uses cloud storage such as Google Drive.	Commonly uses local HDD/SSD storage.
UPDATES	Many system components and apps are updated automatically through the cloud.	The OS and installed applications are generally updated on the device.
INTERNET	Internet is often important for the full experience, although some apps can work offline.	Many installed applications can work without an internet connection.